

Nicolás Moreno Ávila

MAIL nicolaspatricomorenoavila@gmail.com

TEL +56 9 3728 7950

WEB nicolas-moreno.vercel.app

GIT github.com/4mser

FOUNDING ENGINEER · FULL-STACK · AI/ML · INTERACTIVE 3D

PROFESSIONAL SUMMARY

Self-taught software engineer with a background in physics engineering. I build products end to end —architecture, backend, web and native applications, applied machine learning and design— and today I lead engineering at **FIDELYA**, which I built from scratch and which runs with **paying clients**. Before that I led engineering at **GoAuto**, with teams across Chile, LATAM and the United States. Having worked on every layer of the product is what lets me make architectural decisions on technical grounds.

AREAS OF EXPERTISE

Architecture	Backend & infrastructure	Applied AI	Product & design
From the data model to deployment, with the judgement to decide what to build and what to leave alone.	APIs, databases, real time, and the infrastructure that keeps them standing in production.	Computer vision, graph models and agents, put to work on real data.	Design system, interface and copy: what makes someone understand and actually use what was built.

PROFESSIONAL EXPERIENCE

Cofounder, CEO & CTO – FIDELYA

JAN 2025 – PRESENT · SANTIAGO, CHILE · MULTI-TENANT SAAS

- Designed and built a **multi-tenant platform with one isolated database per venue** — NestJS, MongoDB, Next.js and real-time WebSockets — today **105 modules, 123 controllers, 179 data models and 1,103 endpoints** in production.
- Built it **solo from zero**: backend, panel, native apps, infrastructure, design system and brand. Now I lead a team of six and own the architecture.
- Shipped the **white-label mobile app**: one codebase that produces a separate iOS and Android app per venue, published to the App Store and Google Play with over-the-air updates.
- Built **AURA**, the retention engine: it predicts which members are drifting away from real consumption behaviour and acts on it, fed by graph/tensor models and Graph Neural Networks from the in-house research lab (formerly Entropía Technologies).
- Built the parts nobody wants to build: loyalty engine, electronic invoicing, POS and cash register, printing bridge, inventory with recipes, bookings, and a permission system with seven roles.
- Own the release path: I review and merge the team's branches into production, keep the architecture coherent as more people touch it, and onboard each new developer into it.

Cofounder & CTO – B&N Solutions

JUL 2026 – PRESENT · SANTIAGO, CHILE · TECHNOLOGY STUDIO

- Custom software, AI automation and high-end interactive web for clients — including a **real-time Navier-Stokes fluid simulation written in plain WebGL**, with no 3D library, running at 60 fps on a phone.
- Own the whole delivery: discovery, architecture, build, and the documents and decks that sell it.

Tech Lead & Full-Stack Engineer – GoAuto (Dropout Capital)

OCT 2025 – JUL 2026 · REMOTE · LATAM + US TEAM

- Owned architecture and delivery of an AI-powered dealership-management platform running **51 dealerships across Chile**.
- Led a team of five-plus engineers across LATAM and the US, setting architecture and product direction.

Full-Stack & Machine Learning Engineer – Physical AI Lab (Chile)

MAR 2024 – SEP 2024 · REMOTE · ROBOTICS, AI & SIMULATION (PHYSICALAILAB.CL)

- Built a **convolutional neural network** that detects faults on power-transmission lines for SAESA, a Chilean energy utility, from field imagery.
- Shipped production full-stack features alongside the model.

Freelance Full-Stack Developer – Independent

2022 – 2024 · REMOTE

- Landing pages, web automations and small custom systems, end to end — from the conversation with the client to the deploy.

SELECTED PROJECTS

PLATFORM & PRODUCT

FIDELYA – Customer Intelligence OS	Multi-tenant platform for venues: loyalty, menu, table ordering, bookings, invoicing, campaigns and analytics. One isolated database per venue, so a client can be handed their data and leave.	NESTJS · MONGODB · NEXT.JS
Loyalty engine	The core of the business: points per spend, per visit or hybrid, tier ladders, rule-based rewards and badges earned from real behaviour. Configured per venue without touching code.	NESTJS · MONGODB · CRON
Table ordering with a shared cart	Scan the QR printed on your table and order from your phone. If several of you are sitting there, you share one live cart, each adds their own, and the bill is split when you close it.	WEBSOCKETS · REACT NATIVE
Floor app for the staff	The other end of every scan: tables by zone and state, tickets to the kitchen, touch point of sale, split bills and payment, on the waiter's phone.	REACT NATIVE · SOCKETS
White-label native apps	One React Native codebase that builds a separate app per venue — its own name, icon, palette and features. Published to both stores with over-the-air updates.	EXP0 · EAS
Club Happy People	The app of a dinner-theatre venue in Santiago, live on the App Store: points, menu, table ordering, show listing and an in-house party game. From the first commit to the store, alone.	APP STORE · EXP0
Electronic invoicing engine	Chilean tax documents end to end: signing, issuing, tracking and the accounting behind it — built inside the platform, not bolted on.	NODE · CRYPTO · SII
Point of sale, cash register & inventory	Register with open and close, cash counts, inventory with recipes that discount stock per dish, and a printing bridge to USB thermal printers.	NODE · WEBUSB

ARTIFICIAL INTELLIGENCE & COMPUTER VISION

AURA – retention engine	Predicts which members are drifting away from real consumption behaviour and fires the action without anyone pressing a button. Fed by graph and tensor models from the in-house research lab.	GNNS · TENSOR MODELS · PYTHON
AMBAR – post-sale agent	An agent that takes over the conversation after the purchase: follow-up, resolution and recovery, with the customer's context and history in hand.	LLMS · AGENTS
Paper receipts to points – in progress	A neural network that reads a photographed paper receipt and credits the right points, so a venue can join the loyalty programme without integrating its point of sale.	VISION · OCR · PYTORCH
Power-line fault detection	A convolutional network that flags anomalies on transmission lines from field imagery, for a Chilean energy utility (SAESA).	PYTORCH · CNN
Drone point clouds for line inspection	Processing of drone-captured point clouds to inspect power lines in the field: segmentation of the conductors, distance to vegetation and encroachment detection. For SAESA with Akame Ingeniería.	PYTHON · LIDAR
Computer vision on a quadruped robot	Segmentation and anomaly analysis of building windows from what a Spot robot captures, with the control and analytics platform around it: routes, findings and reports.	SEGMENTACIÓN · SPOT

SELECTED PROJECTS (CONT.)

SIMULATION, GRAPHICS & PLAY

WebGL fluid simulation	A Navier-Stokes solver written in plain WebGL — advection, vorticity and a Jacobi pressure solve — with no 3D library. Runs at 60 fps on a phone and reacts to scroll and touch.	WEBGL · GLSL
3D matter sculpted by scroll	A metaball field with analytic normals, its height taken from the square root of the field rather than the raw Gaussian. Each capability of a company is a different simulation that morphs into the previous one as you scroll.	WEBGL · SHADERS
Game engine with memory	A party-game engine that deals from a 357-card deck without repeating, remembers what came out on previous nights, and shifts its own tone as the night goes on.	TYPESCRIPT · REACT NATIVE
Simulation lab	Close to forty interactive pieces — Navier-Stokes, Chladni, Gray-Scott, boids, double slit, Collatz, curvature — written from scratch, as study and as proof.	CANVAS · WEBGL
Interactive documents	Client decks and technical documents that are real web pages: scroll-driven simulations and a letter-format PDF generated from the same source.	GSAP · CANVAS · PUPPETEER

DATA, HARDWARE & INTEGRATIONS

Ticketing integration	The venue's show listing inside the app: cast deduced from the event title, live availability, and purchase without leaving. Their public API leaked bank details — everything is proxied and stripped.	REST · CACHING
Hardware & IoT	A wearable on a Raspberry Pi 5, and lighting control over the local network by speaking a vendor protocol directly — no cloud, no app in the middle.	RASPBERRY PI · PYTHON · LAN
Catalogue importer	Takes the menu a venue already has online and loads it as a working demo in minutes — categories, prices and photos — so they can be shown their own app before the first meeting.	NODE · SCRAPING
Campaigns & per-venue email	Segmented campaigns that go out with each venue's own sender and domain, not the provider's: transactional and campaign email separated per tenant.	RESEND · SEGMENTATION
Bookings & spaces	Table, service and space reservations, with availability per professional, time windows and date blackouts.	NESTJS · CALENDAR
Cross-venue passport	The brand's own app: a map of venues, points that travel between them and a global search, on the same multi-tenant base.	REACT NATIVE · MAPS
Social asset generator	Brand carousels generated from a panoramic canvas cut into slides, with each venue's identity applied.	CANVAS · NODE

TECHNICAL SKILLS

LANGUAGES

 TypeScript

 JavaScript

 Python

 SQL

BACKEND & DATA

 NestJS

 Node.js

 WebSockets

 MongoDB

 PostgreSQL

 Redis

REST

FRONTEND

 Next.js

 React

 Tailwind

Zustand

PWA

MOBILE

 Expo

 React Native

 App Store

 Google Play

EAS Build / OTA

INTERACTIVE & 3D

 WebGL / GLSL

 Three.js

Canvas 2D

 GSAP

custom engines

AI / ML

 PyTorch

 Hugging Face

 OpenCV

 scikit-learn

 NumPy

 pandas

 LLM agents

 Gemini

Groq

graph neural nets

computer vision

AI TOOLING

 Claude Code

 Cursor

custom subagents

agentic workflows

INFRASTRUCTURE

 Fly.io

 Vercel

 AWS S3

 Cloudflare

 Railway

 Docker

 GitHub Actions

CI/CD

 Raspberry Pi

PRODUCT & DESIGN

 Figma

design systems

 Mercado Pago

electronic invoicing

multi-tenant

HOW I WORK

01

Depth first, so I can lead

I have taken products from an empty folder to the App Store on my own — schema, API, panel, app, infrastructure, design and copy. That is exactly what lets me split the work, review it and be accountable for someone else's code.

02

Physics before frameworks

Three years of physics engineering. It shows up where it matters: simulations, models, and knowing when a problem is maths and not more code.

03

Design is not decoration

I build the design system and the interface myself. A product that nobody understands is a product that nobody uses.

EDUCATION

Physics Engineering – Universidad Andrés Bello

2019 – 2022 · THREE YEARS, INCOMPLETE · DATA SCIENCE AND MATHEMATICAL MODELLING

Technical High School – Telecommunications

2014 – 2017 · PROGRAMMING, NETWORKING, ELECTRONICS AND FIBRE OPTICS

LANGUAGES & HONORS

SPANISH **Native**

ENGLISH **Professional working proficiency**

Founder Institute Chile 2026 — selected.

What I am looking for: a technically demanding team where I can get properly involved in the problems — from the architectural decisions through to what actually runs in production. I value autonomy, technical judgement, and working with people who raise the level of the team.